

HDOC DAY-25

Title: Decreasing Pattern Generation using Parameterless Functions and Nested Loops
1. Problem Statement Write a menu-driven C program using a switch statement to print various "Decreasing patterns" selected by the user. Use nested loops without arrays. Each pattern activity must be implemented in a separate parameterless function returning void (void function_name(void)):

1. **Decreasing right-angled triangle:** Input n and print a decreasing right-angled triangle using asterisks.
2. **Decreasing natural-number pattern:** Input n and print natural numbers from 1 to n , reducing count by 1 each row.
3. **Decreasing repeated-number pattern:** Input n , begin with n in row 1, print each value as many times as its value.
4. **Reverse-number triangle:** Input n , print natural numbers in descending order beginning with current row length ending in 1.
5. **Exit:** Terminate program loop.

2. Algorithms

Main Control Flow

1. Display Decreasing Patterns Menu inside a do-while loop.
2. Read choice and route through switch statement to execute corresponding parameterless function.
3. Exit when choice 5 is entered.

Pattern Functions

- **print_decreasing_right_angled():** Read n . Outer loop i from n down to 1, inner loop j from 1 to i : print *.
- **print_decreasing_natural_number():** Read n . Outer loop i from n down to 1, inner loop j from 1 to i : print j .
- **print_decreasing_repeated_number():** Read n . Outer loop i from n down to 1, inner loop j from 1 to i : print i .
- **print_reverse_number_triangle():** Read n . Outer loop i from n down to 1, inner loop j from i down to 1: print j .

3. Test Case Table

Test Case	Pattern / Option	Input n	Expected Pattern Output	Actual Output	Status
1	Decreasing Star Triangle	$n = 4$	Rows: 4 stars / 3 stars / 2 stars / 1 star	Matched expected output	Pass
2	Decreasing Natural Numbers	$n = 4$	Rows: 1 2 3 4 / 1 2 3 / 1 2 / 1	Matched expected output	Pass
3	Decreasing Repeated Numbers	$n = 4$	Rows: 4 4 4 4 / 3 3 3 / 2 2 / 1	Matched expected output	Pass

Test Case	Pattern / Option	Input n	Expected Pattern Output	Actual Output	Status
4	Reverse-Number Triangle	$n = 4$	Rows: 4 3 2 1 / 3 2 1 / 2 1 / 1	Matched expected output	Pass
5	Invalid Choice	Choice = 9	Error message displayed	Error message shown	Pass
6	Exit Option	Choice = 5	"Exiting program ..."	Program exited	Pass

4. Source Code Listing

```
#include <stdio.h>

void print_decreasing_right_angled(void) {
    int n;
    printf("\n--- Decreasing Right-Angled Triangle ---\n");
    printf("Enter number of rows (n): ");
    scanf("%d", &n);
    if (n <= 0) { printf("Error: Input must be a positive integer.\n"); return; }
    for (int i = n; i >= 1; i--) {
```

```

        for (int j = 1; j <= i; j++) printf("*");
        printf("\n");
    }
}

void print_decreasing_natural_number(void) {
    int n;
    printf("\n--- Decreasing Natural-Number Pattern ---\n");
    printf("Enter number of rows (n): ");
    scanf("%d", &n);
    if (n <= 0) { printf("Error: Input must be a positive integer.\n"); return; }
    for (int i = n; i >= 1; i--) {
        for (int j = 1; j <= i; j++) printf("%d ", j);
        printf("\n");
    }
}

void print_decreasing_repeated_number(void) {
    int n;
    printf("\n--- Decreasing Repeated-Number Pattern ---\n");
    printf("Enter number of rows (n): ");
    scanf("%d", &n);
    if (n <= 0) { printf("Error: Input must be a positive integer.\n"); return; }
    for (int i = n; i >= 1; i--) {
        for (int j = 1; j <= i; j++) printf("%d ", i);
        printf("\n");
    }
}

void print_reverse_number_triangle(void) {
    int n;
    printf("\n--- Reverse-Number Triangle ---\n");
    printf("Enter number of rows (n): ");
    scanf("%d", &n);
    if (n <= 0) { printf("Error: Input must be a positive integer.\n"); return; }
    for (int i = n; i >= 1; i--) {
        for (int j = i; j >= 1; j--) printf("%d ", j);
        printf("\n");
    }
}

```

```

int main(void) {
    int choice;
    do {
        printf("\n===== DECREASING PATTERNS MENU
=====\\n");
        printf("1. Decreasing right-angled triangle\\n");
        printf("2. Decreasing natural-number pattern\\n");
        printf("3. Decreasing repeated-number pattern\\n");
        printf("4. Reverse-number triangle\\n");
        printf("5. Exit\\n");
        printf("Enter choice (1-5): ");
        scanf("%d", &choice);

        switch (choice) {
            case 1: print_decreasing_right_angled(); break;
            case 2: print_decreasing_natural_number(); break;
            case 3: print_decreasing_repeated_number(); break;
            case 4: print_reverse_number_triangle(); break;
            case 5: printf("Exiting program...\\n"); break;
            default: printf("Error: Invalid choice!\\n"); break;
        }
    } while (choice != 5);
    return 0;
}

```

4. Execution & Verification Screenshots

```
Session Actions Edit View Help

(kali@kali)-[~]
└─$ mkdir HDOC_Day25
cd HDOC_Day25

(kali@kali)-[~/HDOC_Day25]
└─$ nano day25.c

(kali@kali)-[~/HDOC_Day25]
└─$ gcc day25.c -o day25

(kali@kali)-[~/HDOC_Day25]
└─$ ./day25

===== DECREASING PATTERNS MENU =====
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter choice (1-5): 1

--- Decreasing Right-Angled Triangle ---
Enter number of rows (n): 4
****
***
**
*

===== DECREASING PATTERNS MENU =====
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter choice (1-5): 2

--- Decreasing Natural-Number Pattern ---
Enter number of rows (n): 4
1 2 3 4
1 2 3
1 2
1

===== DECREASING PATTERNS MENU =====
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter choice (1-5): 3

--- Decreasing Repeated-Number Pattern ---
Enter number of rows (n): 4
4 4 4 4
3 3 3
2 2
1

===== DECREASING PATTERNS MENU =====
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter choice (1-5): 4

--- Reverse-Number Triangle ---
Enter number of rows (n): 4
4 3 2 1
3 2 1
2 1
1

===== DECREASING PATTERNS MENU =====
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter choice (1-5): 5
Exiting program ...

(kali@kali)-[~/HDOC_Day25]
└─$
```